DS ASSIGNMENT
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ROLL NUMBER:21011P0529
CSE IDP
SECTION B.

/*Q. Design an appropriate hash function and Write a program to list all unique words in the give of occurrence of each word along with the number of comparisons made for insert search the words

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#include<stdio.h>
#include<stdlib.h>
#include<conio.h>
#include<string.h>
#define maxwords 10000
#define size 26 struct node
{     char data[50];     int
frequency;     int key;
   struct node *next;
}*newnode;
struct node *chain[size]; char
word[50]; int count,p; void init()
{     int i;
   for(i = 1; i <=size; i++)         chain[i] = NULL;
}
int getkey(char temp[50])
{     char first;
first=temp[0];
   if((first=='a')||(first=='A'))
   {return 1;}
   else if((first=='b')||(first=='B'))
   {return 2;}
   else if((first=='c')||(first=='C'))
   {return 3;}
   else if((first=='d')||(first=='D'))
   {return 4;}
   else if((first=='e')||(first=='E'))
   {return 5;}
   else if((first=='f')||(first=='F'))
   {return 6;}
   else if((first=='g')||(first=='G'))
   {return 7;}
   else if((first=='h')||(first=='H'))
   {return 8;}
   else if((first=='i')||(first=='I'))
   {return 9;}
   else if((first=='j')||(first=='J'))
   {return 10;}
   else if((first=='k')||(first=='K'))
   {return 11;}
   else if((first=='l')||(first=='L'))
   {return 12;}
   else if((first=='m')||(first=='M'))
   {return 13;}
   else if((first=='n')||(first=='N'))
   {return 14;}
   else if((first=='o')||(first=='O'))
   {return 15;}
   else if((first=='p')||(first=='P'))
   {return 16;}
   else if((first=='q')||(first=='Q'))
   {return 17;}
   else if((first=='r')||(first=='R'))
   {return 18;}
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   else if((first=='s')||(first=='S'))
   {return 19;}
   else if((first=='t')||(first=='T'))
   {return 20;}
   else if((first=='u')||(first=='U'))
   {return 21;}
   else if((first=='v')||(first=='V'))
   {return 22;}
   else if((first=='w')||(first=='W'))
   {return 23;}
   else if((first=='x')||(first=='X'))
   {return 24;}
   else if((first=='y')||(first=='Y'))
   {return 25;}
   else if((first=='z')||(first=='Z'))
   {return 26;} }
void search(char temp[50])
{     struct node *temp1;     int
key=0,l=0,j=0,length,n=0;
length=strlen(temp);     key=getkey(temp);
temp1=chain[key];     l=0;     while(temp1-
>next)
   {
      for(j=0;temp[j]!='\0';j++)
      {
         if(temp[j]==temp1->data[j])
         {l++;}         }
if(l==length)         break;
n++;     temp1=temp1->next;
   }
      printf("length:%d\n",length);
if(l==length)         {printf("found\n");}         else
      {printf("not found\n");}
      printf("number of comparisions made:%d",n);
}
void insert(char word[50])
{     int key,j;
   newnode=(struct node *)malloc(sizeof(struct node));
for(j=0;word[j]!='\0';j++)     {newnode->data[j]=word[j];}     newnode-
>next =NULL;         key=getkey(word);          if(chain[key] == NULL)
{chain[key] = newnode;}     else
   {struct node *temp = chain[key];
while(temp->next)     {temp = temp->next;}
temp->next = newnode;}
} void print() {
int i;
   for(i = 1; i<=size; i++)
   {
      struct node *temp = chain[i];
printf("chain[%d]->",i);         while(temp)
      {printf("%s[%d]->",temp->data,temp->frequency);         temp =
temp->next;}         printf("NULL\n");         printf("\n");         printf("\n");
   } }
int main()
{
   char value[50];
   int i,isunique,k,l,index=0,p=0,r=0;     char
words[maxwords][50],word[50],temp[50];     int
count[maxwords];     FILE *fp;
   fp=fopen("data.cpp.cpp","r");
for(i=0;i<maxwords;i++)
   {count[i]=0;}
   while(fscanf(fp, "%s", word)!=EOF)
```

```
    {
        strlwr(word);        l=strlen(word);
        if((word[l-1]=='.')||(word[l-1]==',')||(word[l-1]=='!')||(word[l-1]=='?')||(word[l-1]==';')||(word[
        {word[l-1]='\0';}
        if((word[0]=='.')||(word[0]==',')||(word[0]=='!')||(word[0]=='?')||(word[0]==';')||(word[0]=='")||
        {word[0]='\0';}        isunique=1;
        for(i=0;i<index && isunique;i++)
{k=strcmp(words[i],word);        if(k==0){isunique=0;}}
if(isunique==1)
        {strcpy(words[index],word);
count[index]++;        index++;}        else
        {count[i-1]++;}
    }
    for(p=0;p<index;p++)     {insert(words[p]);
    newnode->frequency=count[p];}
print();     fclose(fp);     search("won");
return 0;
}
```

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chain[1]->and(83)->alder(1)->ahead(1)->a(62)->always(3)->above(1)->another(4)->away(3)->armed(2)->any(3)->arising(1)->all(8)->as(26)->asked(6)->an(6)->accomplished(1)->at(14)->adding(2)->act(1)->are(1)->ability(
1)->added:(1)->able(1)->again(2)->appearing(1)->affected(1)->ascending(1)->advance(1)->astonished(1)->all,--as(1)->ain't(4)->appreciation(1)->all-pervading(1)->astonishment(1)->appeared(1)->assured(1)->about(2)-
>against(2)->ample(2)->average(1)->affiliated(1)->active(2)->assented(1)->acquaintance(4)->answered(1)->alone(1)->assurance(1)->anger(1)->adjective(1)->among(1)->assuming(1)->already(4)->am(1)->admire(1)->accoun
t(1)->after(1)->adorned(1)->aside(1)->acquiescence(1)->ablutions(1)->awe(1)->arrant(1)->archery(1)->almost(2)->aware(1)->added(1)->actually(1)->adventure(1)->auditors(1)->alleged(1)->arose(1)->absence(1)->answer
(1)->NULL


chain[2]->by{(12)->bret(1)->but(20)->behind(1)->business(1)->buckeye(2)->bushes(2)->black(1)->be(8)->better(2)->between(5)->beside(2)->beyond(2)->borrow(1)->beast(2)->before(6)->boys(1)->being(1)->been(4)->blank
(1)->blinding(1)->below(4)->bowlder(1)->beginning(3)->books(1)->best?(1)->best(1)->beguiled(1)->become(1)->bring(1)->both(1)->boys,(1)->back(1)->boys,"--what(1)->began(1)->barking(1)->brought(1)->briefly(1)->bla
ndly(1)->beauty(1)->bewildered(1)->battery(1)->boyish(1)->bridge(1)->boards(1)->breath(1)->breaking(1)->broken(1)->NULL


chain[3]->clear(1)->california(2)->classical(1)->circumspect(1)->contemplation(2)->come(1)->communication(1)->crackling(1)->cold(1)->commonplace(1)->course.(1)->came(3)->curiously(1)->caring(1)->can(3)->carried(
2)->continued(1)->could(14)->croupe(1)->chance(1)->conversation(2)->curiosity(1)->chimisal(2)->canyon(1)->cast(1)->celebrity(1)->course(2)->considerable(1)->certainly(3)->count(2)->criticism(1)->considered(1)->c
lothed(1)->considerably(2)->change(1)->contents(1)->companion(5)->chosen(1)->called(1)->conferring(1)->checking(1)->correct(1)->circle(1)->complimentary(1)->certain(1)->call(1)->cottage(1)->commented(1)->care(2)
->commands(1)->cool(2)->cousin(1)->charms(1)->completely(1)->cynical(2)->clear-headed(1)->covering.(1)->cheeks(1)->ceremony(1)->children(1)->coolly(2)->chatted(1)->cavalcade(1)->country(1)->cherokee(1)->calavera
s(2)->cursed(1)->NULL


chain[4]->down(5)->darkening(2)->distrust(1)->discovery(1)->dozen(1)->dared(1)->doubt(1)->deprived(1)->disremember(1)->depends(1)->dragoon(1)->do(3)->disposal(1)->double(2)->done(1)->duty(1)->divide(1)->disposed
(1)->drifted(1)->drew(2)->dickens(4)->darwin(1)->did(5)->dwell(1)->diligent(1)->derringer(1)->diversion(1)->dialect(1)->dignity(1)->don't(6)->d.'(1)->domestic(1)->descend(1)->distance(1)->dogs(1)->door(1)->do--k
earney's(1)->doubtless(1)->daughter's(1)->den(1)->drawl(1)->deliciously(1)->dazed(1)->delivered(1)->delicious(1)->defiance(1)->dear(1)->darkness(1)->disturbed(1)->distanced(1)->NULL


chain[5]->equipped(1)->eyes(4)->eagerness(1)->every(1)->either(1)->en(1)->entirely(1)->embarrassment(3)->equaled(1)->exuberance(1)->element(1)->emotional(1)->enthusiasm(2)->evidently(2)->eccentric(1)->eastern(1)
->ever(1)->etc.(1)->etc(1)->ef(2)->entree(1)->even(1)->easy(2)->enough(1)->evident(1)->excused(1)->ear-shot(1)->equally(1)->excitement(2)->enrapt(1)->emptying(1)->exciting(1)->explained(1)->each(1)->early(1)->en
thusiastic(1)->empty(1)->NULL


chain[6]->friend(5)->fringe(1)->for(13)->first(2)->feet(1)->five(1)->from(9)->faring(1)->fact(6)->fourthly(1)->fifthly(1)->facts(1)->found(2)->feature(1)->filling(1)->finally(1)->few(3)->face(1)->familiarity(1)-
>flank(1)->file(1)->far(1)->forget(1)->fervor(1)->fiction(1)->felicity(1)->fitness(1)->flask(1)->firmly(1)->fastidiousness(1)->family(3)->familiarly(1)->feared(1)->favor(1)->feel(1)->felt(1)->feeling(1)->forgott
en(2)->fetched(1)->forcible(1)->fine(1)->forgive(1)->find(1)->flirt(1)->friend?(1)->foreign(1)->fear(1)->fascinated(1)->father(1)->fair(1)->finding(1)->friends(1)->freaks(1)->fellow(1)->NULL


chain[7]->gullied(1)->geographical(1)->grade(1)->gray(1)->got(3)->good(5)->going(1)->gives(1)->give(1)->gulch(1)->giving(1)->glare(1)->gulfs(1)->gigantic(1)->growing(2)->grasped(1)->great(1)->gratified(1)->girls
,--i(1)->gentle(3)->general(1)->gentlemanly(1)->giddy(1)->girl(1)->glad(1)->girls(2)->gasping(1)->gather(1)->guest(1)->NULL


chain[8]->harte(1)->horse(8)->here(4)->he(59)->had(22)->his(43)->holding(1)->hundred(1)->have(7)->height(1)->halted(2)->hours(3)->half(2)->him(16)->how(3)->hand(2)->holster(1)->heavytree(1)->hill(1)->has(1)->hel
p(3)->heat(1)->hawk(1)->hope(2)->halt(1)->hate(1)->huxley(1)->hesitatingly:--(1)->hearty(1)->heard(1)->humor(2)->homespun(1)->himself(2)->horses(1)->hev(1)->hesitated(3)->habits(1)->hands(1)->hospitality(1)->her
e,--in(1)->heerd(1)->half-aloud(1)->horror(1)->helpless(1)->heaven(2)->hospitable(1)->host(2)->hearing(1)->however(1)->her(5)->hang(1)->he?(1)->highness(1)->him,(1)->honestly(1)->hero(1)->heart(1)->hung(1)->hear
(1)->horses'(1)->hoofs(1)->homicidal(1)->hunting(1)->hasn't(1)->NULL


chain[9]->i(92)->in(32)->impatiently(3)->inaccessible(1)->inquiringly(1)->i'll(5)->indian(4)->it(15)->if(6)->its(5)->idea(1)->into(2)->intelligent(1)->illustration(2)->is(10)->influence(1)->insisted(1)->instantl
y(1)->it?(1)->ideas(1)->it's(4)->introduction(2)->i've(1)->is.(1)->interrupted(1)->inquest(1)->introduced(1)->innocence(1)->interest(2)->isn't(2)->irony(1)->incognito(1)->inform(1)->introduce(2)->incompatible(1)
->indeed(1)->indeed!(1)->idle(1)->intoxicated(1)->inquiry(1)->import(1)->i'd(1)->impression(1)->NULL
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chain[10]->jine(1)->just(5)->jiss(1)->justify(1)->jones(1)->jealous(1)->jingling(1)->jack(1)->NULL


chain[11]->knowledge(1)->kept(3)->knew(3)->kin(1)->known(1)->know(2)->kearney(8)->kate(2)->knows(1)->NULL


chain[12]->loud(1)->looked(7)->little(11)->ledge(2)->looking(1)->locality,--lonely(1)->lead(1)->long(4)->lagrange.(1)->lagrange(3)->laughingly(1)->lived(1)->let(1)->leathern(1)->lift(1)->lazily(1)->level(1)->led
(1)->liked(2)->local(1)->like(3)->launched(1)->language(1)->literature(1)->lyrical(1)->locality(1)->lost(1)->lonely(1)->lose(1)->large(1)->less(1)->lights(1)->lone(2)->let's(1)->leave(1)->lay(2)->lingered(1)->la
nguid(2)->lovely(1)->lower(1)->lovely?(1)->laughed(1)->later(1)->lips(1)->low(1)->learn(1)->NULL


chain[13]->my¿(40)->meant(1)->me(27)->man(15)->mountaineer(1)->miners(1)->mounted(1)->mind(4)->mistaken(1)->medium(1)->moments(1)->mere(1)->much(3)->make(4)->many(1)->mustang(1)->myself(4)->moving(1)->mountain(1
)->made(1)->might(2)->mildly(1)->more(2)->mean--(1)->moment(2)->master(1)->manner(1)->misanthrope(1)->means(1)->miner(1)->mr(3)->mysterious(4)->matter(2)->mind.(1)->most(1)->me!(1)->man's(1)->musingly(1)->me,(2)
->man.(1)->mystification(1)->mean(1)->maniac(1)->may(6)->meanwhile(1)->me.(1)->mingled(1)->maidenly(1)->managed(1)->must(1)->morganatic.(1)->may--she(1)->middle(1)->mexican(1)->mixed(1)->midnight(1)->matter?(1)-
>morning(1)->NULL


chain[14]->not(15)->narrow(1)->no(8)->noted(2)->new(3)->nevertheless(3)->nervous(2)->name.(1)->natural(1)->now(3)->nothing(3)->neither(1)->narrowness(1)->novelist(1)->never(2)->nor(3)->noticing(2)->necessary(1)-
>night(2)->name(7)->named(1)->nature(1)->narrative(1)->next(1)->nobody(1)->noise(1)->NULL


chain[15]->one(8)->of(81)->on(27)->own(7)->other(5)->ordinary(1)->or(10)->one's(1)->odd(1)->over(5)->only(8)->out(7)->offensive(1)->objected(1)->our(3)->oddly(1)->occasion(1)->obliged(1)->olive(1)->often(2)->opi
nion(1)->old(3)->offered(1)->offices(1)->obscure(1)->onnoticed(1)->opposition(2)->object(2)->off(1)->o(1)->oppose(1)->offering(1)->on.(1)->oneasy(1)->observant(1)->NULL


chain[16]->penetrating(1)->perceived(1)->parallel(1)->passed(1)->pouch(2)->plunged(1)->pair(1)->pipe(1)->puffed(1)->puffs(1)->particularly(1)->perfunctory(1)->politeness(1)->pretty(3)->pan(1)->pasear(3)->perhaps
(1)->prolonged(1)->proceed(1)->pines(1)->push(1)->patiently(1)->possible(1)->phlegm(1)->preface(1)->power(1)->pathos(3)->poetry(1)->place(2)->period(2)->precision(1)->poets(1)->professional(1)->pack(2)->pressed(
1)->pursuing(1)->pointed(3)->preferred(1)->politely(1)->proposed(1)->picturesque(1)->proprietor(1)->people(1)->put(1)->play(1)->personal(1)->prepossessing(1)->personally(1)->puzzled(1)->picked(2)->placated(1)->p
racticing(1)->pitiless(1)->perfectly(1)->potentate(1)->pleased(1)->properly(1)->pertly(1)->prejudiced(1)->passing(1)->pink(1)->poor(2)->philosopher(1)->pottage(1)->partitions(1)->praises(1)->personage(1)->playfu
l(1)->peculiar(1)->posse(1)->NULL


chain[17]->quiet(4)->quite(5)->quietly(4)->question(1)->quotation(1)->questions(1)->queer(1)->quietly:(1)->quotes(1)->quitted(1)->NULL


chain[18]->regular(1)->road(2)->rapidly(2)->returned(2)->right(3)->ride(1)->ridden(1)->rider,--(1)->ranged(1)->rather(3)->replied(6)->remember(1)->rebuff(1)->riffle.(1)->rest(2)->revolver(1)->riffle(1)->reck'nin
'(1)->reflection(1)->room(3)->red(2)->rocky(1)->reflected(1)->rising(1)->regularly(1)->reappear(1)->reappearances(1)->referred(1)->read(1)->rough(1)->rarely(1)->rustic(1)->readjusted(1)->route(1)->resumed(2)->re
ckon(1)->recluse(1)->rail(1)->ranch,(1)->respected(1)->respect(1)->repeating(1)->road--i'll(1)->reserve,--known(1)->risks(1)->recognizances(1)->resist(1)->rancho(2)->responsible(1)->reckoned(1)->real(2)->ran(1)-
>reserved(1)->rivalry(1)->romantic(1)->royal(1)->regarding(1)->reason(1)->reader(1)->revelation(1)->retired(1)->retiring(1)->rode(1)->raised(1)->rest.(1)->NULL


chain[19]->slope(1)->stranger!(1)->small(1)->secondly(1)->stranger's(3)->superior(1)->self(1)->salutation(1)->signified(1)->slide.(1)->slide!(1)->second(1)->seen(2)->so(7)->sharp(1)->shower(1)->stones(1)->side(5
)->should(4)->scarcely(1)->size(1)->suggested(1)->surprised(2)->said(13)->sedative(1)->said:(1)->spring(4)->somehow(1)->sorter(1)->such(2)->strong(1)->suggestion(3)->speech(2)->save(1)->surprise(2)->so,(2)->spar
e.(1)->smiling(1)->stranger(5)->slowly(2)->satisfy(1)->silence(1)->sun(1)->shadow(1)->superiority(1)->some(8)->slouch(3)->sympathy(1)->student(1)->staggered(1)->slouching(1)->slangy(1)->sometimes(1)->struck(1)->
softened(2)->saddle(1)->skill(1)->sharing(1)->silver-mounted(1)->say(4)->see(4)->stand(1)->strange(1)->stopping(2)->secluded(1)->sierras(2)->spot(1)->sylvester(13)->seclusion(1)->shadows(1)->suddenly(2)->stop(1)
->sylvester's.(1)->sullenly(1)->slight(1)->sylvester's--why--i(1)->stated(1)->strongly(1)->short(2)->same(2)->shame(1)->solitary(1)->street(1)->superb(1)->self-confidence(1)->smith(1)->strike(1)->soi-disant(1)->
sylvester's(1)->supervise(1)->subjects(1)->submission(1)->spectacle(1)->sense(1)->slang(1)->sitting-room(1)->self-styled(1)->seated(1)->sofa(1)->sitting(1)->seemed(2)->she(2)->supper(1)->sat(1)->severe(1)->submi
t(1)->sex(1)->story(1)->stopped(1)->stable,(1)->statement(1)->slightest(1)->sweet(1)->speaking(1)->saw(1)->silly(1)->self-asked(1)->startled(1)->sound(1)->spurs(1)->shot(1)->sheriff(2)->softly(1)->NULL
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chain[20]->the(162)->trail(12)->this(9)->time(5)->thirty(1)->through(5)->things(2)->to(64)->teamsters(1)->that(48)->traveler(1)->thirdly(1)->timidity(1)->these(2)->tobacco?(1)->times(1)->there(6)->there,(1)->the
n(4)->thought(3)->took(2)->to--(1)->taking(2)->this,(1)->toiled(1)->those(2)->thinking(1)->them(1)->time.(1)->true(1)->toward(1)->tried(1)->transferred(1)->talk(1)->trail's(1)->takin'(1)->through.(1)->their(3)->
twixt(1)->thar(1)->time--but(1)->that's(3)->they(2)->twinkled(1)->turned(1)->thar's(1)->title(1)->taste(1)->touch(1)->traveling(1)->then,(1)->too.(1)->than(1)->train(1)->trail,(1)->trifle(1)->table(1)->trace(1)-
>tears(2)->thin(1)->two(2)->tone(1)->told(1)->too(1)->NULL


chain[21]->up(6)->upon(7)->unknown(3)->underbrush(1)->under(2)->unconsciously(1)->using(1)->us(3)->unlike(2)->usual(1)->undress(1)->unarmed(1)->unrecognizable(1)->up--i(1)->upstairs(1)->unaffected(1)->utterly(1)
->used(1)->unassuming.(1)->NULL


chain[22]->voice(2)->vainly(2)->vocative(1)->various(1)->valley(2)->valley,(2)->views(1)->verb(1)->voluble(1)->NULL


chain[23]->who(4)->was(58)->with(36)->well(3)->went(2)->when(9)->whence(1)->where(3)->whether(2)->wanted(2)->wasn't(1)->what(8)->why?(1)->we(6)->were(6)->way(2)->westering(1)->walls(1)->which(6)->weird(1)->wings
(1)->weary(1)->waiting(1)->would(5)->whisky(1)->will(3)->world(2)->without(1)->whom(1)->whar(1)->why(4)->won(1)->wouldn't(1)->whose(1)->wanderings(1)->while(1)->whisper(1)->words(1)->world-sick(1)->wild(1)->witt
y.(1)->witty(2)->woman(1)->window(1)->NULL


chain[24]->NULL


chain[25]->ye(4)->yards(1)->your(8)->you(16)->yet(5)->you.(1)->you've(1)->you're(1)->yer(3)->yar,(1)->you'd(1)->young(1)->you?(1)->NULL


chain[26]->NULL


length:3
found
number of comparisions made:2
--------------------------------
Process exited after 0.1883 seconds with return value 0
Press any key to continue . . .
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